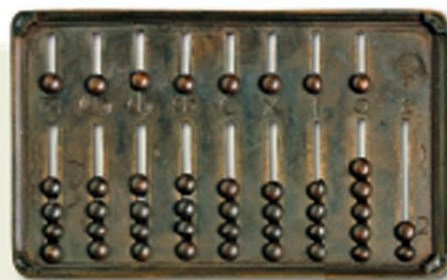


Calculating machines

The word “calculate” comes from the Latin word *calculus* meaning “little stone,” because pebbles were used in ancient times as a counting aid. Then someone had the idea of putting the stones onto a frame, and the abacus was born. Advances in mathematics and astronomy led to the development of the first calculating machines in the 1700s, which evolved into the tablet computers and smartphones we use today.



Replica of a Roman abacus

Early handheld device

The Romans developed the first portable calculating device—a handheld abacus. Made of bronze, it worked by sliding grooved beads up and down the numbered slots, and was probably used by engineers, merchants, and tax collectors, who needed to make on-the-spot calculations.

Female calculators

Until recently, calculating machines were mostly used in business and finance offices. Humans remained better at some complex mathematical tasks. At a time when female employment was rare, the “Harvard Computers” were a team of women mathematicians employed to analyze astronomical data at the Harvard College Observatory, Cambridge, Massachusetts.



The “Harvard Computers” at work, c 1890



Replica of Schickard's calculating machine

Calculating machine

It is believed that German mathematician and astronomer Wilhelm Schickard built the first calculating machine around 1623. He described it in a letter to fellow German astronomer Johannes Kepler, but it was unfortunately lost in a fire. It seems to have combined Napier's bones (see p.74) with toothed wheels for adding and subtracting.

Babbage's Difference Engine

In the 1820s, English mathematician Charles Babbage developed the first of his calculating machines. Its purpose was to create numerical tables without human error. This is a demonstration model, built in 1832, based on Babbage's designs. He never completed the engine himself. Babbage improved on the calculating machine when he later went on to design the Analytical Engine—an early computer.

Key events

c 2700 BCE

The first abacus was probably invented in Sumer (modern-day Iraq). It was soon adopted everywhere, and remains in use in some parts of the world even today.

1622

English mathematician William Oughtred invented the slide rule. Slide rules are used to multiply, divide, and calculate roots and logarithms. You cannot use them to add or take away.

1642

French philosopher and mathematician Blaise Pascal built the first surviving mechanical calculator to be put to practical use. It is known as a Pascaline.

1820

Charles Xavier Thomas of Colmar, France, built a mechanical calculating machine, the arithmometer, for office use. It was a commercial success.

Arithmometer, c 1890

